Journal of Electromyography and Kinesiology 24 (2014) 37–45

Contents lists available at ScienceDirect

Journal of Electromyography and Kinesiology

journal homepage: www.elsevier.com/locate/jelekin

Spatial reorganisation of muscle activity correlates with change in tangential force variability during isometric contractions

Christian A. Mista, Sauro E. Salomoni, Thomas Graven-Nielsen *

Laboratory for Musculoskeletal Pain and Motor Control, Center for Sensory-Motor Interaction (SMI), Department of Health Science and Technology, Faculty of Medicine, Aalborg University, Denmark

ARTICLE INFO

Article history:
Received 11 April 2013
Received in revised form 12 September 2013
Accepted 24 October 2013

Keywords:
Isometric force
High-density EMG
Muscle adaptation
Force variability
Three-dimensional feedback

ABSTRACT

The aim of this study was to quantify the effects of spatial reorganisation of muscle activity on task-related and tangential components of force variability during sustained contractions. Three-dimensional forces were measured from isometric elbow flexion during submaximal contractions (50 s, 5–50% of maximal voluntary contraction (MVC)) and total excursion of the centre of pressure was extracted. Spatial electromyographic (EMG) activity was recorded from the biceps brachii muscle. The centroids of the root mean square (RMS) EMG and normalised mutual information (NMI) maps were computed to assess spatial muscle activity and spatial relationship between EMG and task-related force variability, respectively. Result showed that difference between the position of the centroids at the beginning and at the end of the contraction of the RMS EMG and the NMI maps were different in the medial–lateral direction ($P < 0.05$), reflecting that muscle regions modulate their activity without necessarily modulating the contribution to the task-related force variability over time. Moreover, this difference between shifts of the centroids was positively correlated with the total excursion of the centre of pressure at the higher levels of contractions (>30% MVC, $R^2 > 0.30$, $P < 0.05$), suggesting that changes in spatial muscle activity could impact on the modulation of tangential forces. Therefore, within-muscle adaptations do not necessarily increase force variability, and this interaction can be quantified by analysing the RMS EMG and the NMI map centroids.
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1. Introduction

During sustained contractions the muscle activation strategy changes in order to maintain the same force output (Farina et al., 2008). The reorganised muscle activity exposes neural flexibility of motor unit recruitment, suggesting that peripheral and central mechanisms modulate the motor units activation during sustained contractions (Gandevia, 2001). Moreover, new evidence suggests that fatigue effects are uneven distributed over the muscle (Gallina et al., 2011; Hedayatpour et al., 2008), reflecting spatial differences of the motor unit control strategies within the muscle (Farina et al., 2008). The changes in the muscle activity can alter the force output direction (Tucker and Hodges, 2010), leading to modulations of the force variability (Salomoni and Graven-Nielsen, 2012a). Therefore, spatial reorganisation of the muscle activity distribution could yield to modulation of the force variability.

Several reports have tried to associate the changes in motor unit recruitment strategies with changes in force output and its variability (Holtermann et al., 2009; Tucker and Hodges, 2010; Yao et al., 2000). Dissimilarity of motor unit synchronisation behaviour in different locations of the muscle has been positively associated with the force variability (Holtermann et al., 2009). However, several studies have found inconsistencies between motor unit synchronisation and force variability, showing that the contribution of motor unit synchronisation on the force variability is still unresolved (Semmler, 2002). Recent findings suggest that recruitment of additional motor units has a stronger impact in force variability than motor unit synchronisation (Contessa et al., 2009).

Previous studies have focused on unidirectional force variability (Bandholm et al., 2008; Missenard et al., 2009) and recent investigations included three-dimensional force recordings obtaining additional information regarding motor control performance analysing the variability of tangential forces (Salomoni and Graven-Nielsen, 2012a, 2012b; Svendsen and Madeleine, 2010). The muscle contribution to the force output can be estimate by surface electromyography, which is a non-invasive technique assessing global activity of motor unit populations (Disselhorst-Klug et al., 2009; Farina et al., 2004). Spatial information of the

* Corresponding author. Address: Laboratory for Musculoskeletal Pain and Motor Control, Center for Sensory-Motor Interaction (SMI), Department of Health Science and Technology, Faculty of Medicine, Aalborg University, Fredrik BajersVej 7D-3, 9220 Aalborg E, Denmark. Tel.: +45 9940 9832; fax: +45 9815 4008.
E-mail address: tgn@hst.aau.dk (T. Graven-Nielsen).
URL: http://www.smi.hst.aau.dk/~tgn (T. Graven-Nielsen).

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http://dx.doi.org/10.1016/j.jelekin.2013.10.014

myoelectric activity can be observed by recording activity from multiple near locations over the same muscle. This spatiotemporal information of muscle activity allows the assessment of adaptations within a muscle (Falla and Farina, 2007; Farina et al., 2008; Gallina et al., 2011; Holtermann et al., 2009).

It is hypothesised that spatial reorganisation of muscle activity modulates force variability during sustained contractions. To test this hypothesis, this study quantifies the relationship between spatial reorganisation of the biceps brachii muscle and task-related/tangential force variability during isometric elbow flexion contractions using linear and nonlinear techniques (Farfán et al., 2010; Madeleine et al., 2011).

2. Methods

2.1. Subjects

Fourteen right-handed subjects (8 males, age 26.6 ± 2.8 years; height 170.2 ± 8.1 cm; weight 68.3 ± 11.9 kg, mean ± SD) were included in the study. Subjects were free of upper limb pain, and they had no history of orthopaedic disorders affecting the upper limb region and no history of neurological disorders. All subjects received written and oral description of the procedures and gave an informed consent consistent with the Declaration of Helsinki ethical standards, and the experimental procedures were approved by the local ethics committee (N-20110079).

2.2. Experimental protocol

This study was conducted in a single session where subjects sat upright in a neutral position with their back resting against a height-adjustable chair. The elbow joint of the right arm was flexed at 90° while the forearm was in supinated position, and the wrist was in slight contact with a three-dimensional force transducer, which recorded the force output during elbow flexion (Fig. 1a).

The experiment consisted of isometric submaximal elbow-flexion contractions denominated "three-dimensional" contractions, since subjects were asked to match three force components at 5%, 15%, 30%, and 50% MVC for 50 s. Before the three-dimensional contractions, MVC was recorded by performing two consecutive maximal contractions for 10 s with an interval of 60 s. Thereafter, a set of submaximal contractions 50 s long with 3 min in-between were assessed. In these contractions, only the task-related force component (Fig. 1a: *Z* direction) was displayed on the visual feedback, while medial–lateral (Fig. 1a: *Y* direction) and anterior–posterior (Fig. 1a: *X* direction) force components were recorded. The average magnitudes of the tangential forces recorded were used to set the dimension and position of the target force in the three-dimensional contractions. Finally, the three-dimensional contractions were performed. A continuous visual feedback of the resultant force output (Fig. 1b: circle) was given, as well as the three-dimensional visual force target (Fig. 1b: square). Subjects performed 2 trials of each set of submaximal contractions levels in random order. Only the three-dimensional contractions were used in further analyses. During all the submaximal contractions, the visual feedback showed a ramp-and-hold contraction with 4 s of ramp phase from repose position to the required level of contraction. All motor tasks were followed by 5 min of rest, where subjects were able to move their arm if needed.

2.3. Visual feedback

The visual feedback was presented through a two-dimensional plot on a computer screen. Force components parallel with the axial plane of the wrist were represented as a moving black open circle. The open circle position on the *Z* axis represented the task-related force direction, while the position on the *Y* axis represented medial–lateral force directions. The remaining *X* axis was represented by a circle, concentric to the black open circle, showed in red or blue colour to indicate anterior or posterior error of the force output relative to the target force (Fig. 1b). Moreover, the concentric circle radius represented the magnitude of the deviations from the anterior–posterior force target. The force target on the *Z* and *Y* axis was represented by a square, which moved on the screen over time during the ramp phase. The size of the square was set to 5% of the *Z* and *Y* axes scales for each level of contraction. To fulfil the required motor task, the circle should be inside the square throughout the whole task. The final position of this target was displayed by a dashed square.

2.4. Force recordings

The three dimensional force components and torques were measured using a six-axis load cell transducer (MC3A 250, AMTI,

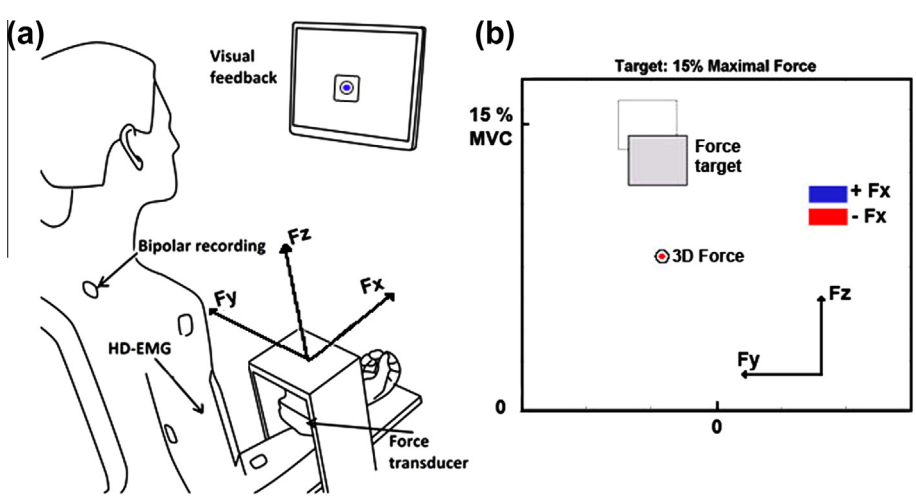

Fig. 1. (a) Schematic illustration of the setup. The EMG electrode grid was placed on the biceps brachii muscle, and bipolar electrodes were located on relevant elbow-flexion muscles. Force output was recorded in the task-related (*Z*) and the tangential (*X* and *Y*) directions using a three-dimensional force transducer. (b) Example of the visual feedback provided to subjects. The target force was represented by a moving square in the Fy-Fz plane, and the final position was shown as an additional dashed square. The continuous three-dimensional force output was represented by a black circumference, which moved in the Fy-Fz plane, and a circle which changes colour and size depending on the magnitude and direction on the Fx direction. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

USA) with high sensitivity (0.054, 0.054, 0.0134 V/N for Fx, Fy, Fz; and 2.744, 2.744, 2.124 V/N m for Mx, My, Mz). The analogue output of the transducer was amplified, and low-pass filtered at 1 kHz (MSA-6, AMTI, USA). The force and torque signals were sampled at 2 kHz and stored after 12 bits A/D conversion.

2.5. Bipolar electromyography recordings

Bipolar surface electromyography (EMG) signals were recorded from the right brachioradialis, triceps brachii, trapezius, and deltoid muscles using pairs of disposable Ag/AgCl surface electrodes (Ambu Neuroline 720, Denmark). The electrodes were placed 2 cm apart and positioned in accordance with the SENIAM recommendations. A ground electrode was placed around the participant's right wrist. EMG signals were amplified (Counterpoint MK2, Dantec, Denmark), band-pass filtered (10–1000 Hz), sampled at 2 kHz, and stored after 12 bits A/D conversion.

2.6. High-density electromyography recordings

Myoelectric signals were recorded using an adhesive grid of 64 electrodes (model ELSCH064R3S, OTBioelectronica, Italy), consisting of 13 rows and 5 columns of electrodes (1-mm diameter, 8-mm inter-electrode distance in both directions) with the lack of one electrode in the upper right corner. The electrode grid was placed over the muscle belly of the right biceps brachii, and grid's columns were in parallel to the humerus bone (Fig. 1a). To assure comparable position of the grid for all subjects, the grid midpoint was aligned with the two biceps heads partition, assessed by palpation, and the bottom of the grid was located 2 cm from the elbow joint (anticubital fossa). The grid was attached using an adhesive foam (KITAD064, OTBioelectronica, Italy) leaving small cavities between silver–silver chloride electrode surfaces and the skin area, which were filled with conductive cream (AC Cream, OTBioelectronica, Italy). Myoelectric signals from the grid were recorded in bipolar configuration, amplified (64-channel surface EMG amplifier, SEA64, OTBioelectronica, Torino, Italy), filtered (3 dB bandwidth, 10–500 Hz), sampled at 2048 Hz, and converted to 12-bit digital samples.

2.7. Data analysis

Force and torque signals were digitally low-pass filtered at 20 Hz using a second order Butterworth filter. EMG signals were digitally band-pass filtered (Butterworth, second order, frequency bandwidth 20–500 Hz) and full-wave rectified. Data analysis was performed on 8 s epochs in three time intervals during the steady phase of the contractions (8–16 s, 24–32 s, 40–48 s), excluding the ramp phase of the contractions in the analysis. The epochs were selected to characterise adaptations of the force output and the muscle activity over time.

Force coefficient of variation (CV = SD/mean force) was calculated to evaluate force variability in the task-related direction of the contraction, and tangential forces variability was assessed using the total excursion of the centre of pressure (CoP) (Salomoni and Graven-Nielsen, 2012a, 2012b). Furthermore, root mean square (RMS) EMG was assessed for each bipolar recording and for each channel in the grid during these epochs. The average RMS EMG from all channels on the grid was used as a parameter of muscle activation.

The rectified EMG signals were smoothed by applying a low-pass filtering (Butterworth, second order, frequency cut-off 30 Hz), after which the normalised mutual information (NMI) was assessed between task-related force (Fz) and each EMG signal using 128 bins, including all grid channels and bipolar recordings. EMG signals were compensated with 40 ms (Corcos et al., 1992) for the electromechanical delay. The MI between force and EMG signals illustrates information shared between these two signals (Brown et al., 2012). The MI between two signals x and y is defined as

$$MI(x,y) = \sum_{xi,yj} Pxy(xi,yj) * log_2 \frac{Pxy(xi,xj)}{Px(xi) * Py(yj)}$$

where Px and Py are the probability that a recorded value will find the system in the ith element of the bin, and Pxy is the joint probability density for x and y. The MI always shows non-negative values and is null when both signals are stochastically independent. Strehl and Ghosh (2003) proposed a MI normalisation defined by

$$NMI(x,y) = \frac{MI(x,y)}{\sqrt{H(x) * H(y)}}$$

where $H(x)$ and $H(y)$ represents the entropy of the signals x and y, respectively.

RMS EMG maps of the grid were extracted to assess spatial activation of the biceps brachii muscle. In addition, NMI maps of EMG activity and the task-related force were developed in order to assess the spatial contribution of the muscle activation on the force variability. RMS EMG and NMI centroid coordinates of the maps were calculated to describe muscle adaptation, defined as Gx and Gy for the medial–lateral and cranial–caudal direction of the map, respectively (Madeleine et al., 2006). Furthermore, centroid shifts were assessed as the absolute difference between the initial and the final positions of the centroid coordinates (Farina et al., 2008). The difference between centroids' shifts of the RMS EMG and NMI maps were calculated and interpreted as difference between the distribution of EMG activity and its contribution to the force variability.

2.8. Statistical analysis

The distribution of the CoP parameter was skewed and thus it was logarithmically transformed in order to normalise the distribution. Force and CoP were assessed using two way repeated-measures analysis of variance (RM-ANOVA) with *Contraction level* (5%, 15%, 30%, and 50% of the MVC force) and *Time* (beginning, middle, and end) as within-subject factors. Similarly, RMS EMG and NMI of each bipolar recording and grid averaged values were assessed using RM-ANOVA with the same factors (*Contraction level* and *Time*). In addition, centroid shifts were assessed using a two way RM-ANOVA with *Contraction level* and *Type of map* (RMS and NMI) as within-subject factors. Since their anatomical and physiological difference, the medial–lateral and cranial–caudal direction of the maps were analysed independently. Significance was accepted for P-values lowers than 0.05 and Student–Newman–Keuls (NK) post hoc tests were applied when it was required. In addition, Pearson's correlation was used to investigate the correlation of difference between maps' centroids shifts and the total excursion of the CoP in the tangential force output. Data are presented as mean values and standard error of the mean (SEM).

3. Results

Representative signals recorded during elbow flexion at 50% MVC are shown in Fig. 2, showing changes in muscle activation characteristics during the 50 s contractions. The biceps brachii RMS EMG map was characterised by two main zones of activity and a low activity zone, associated with the innervation of the muscle (Fig. 2c).

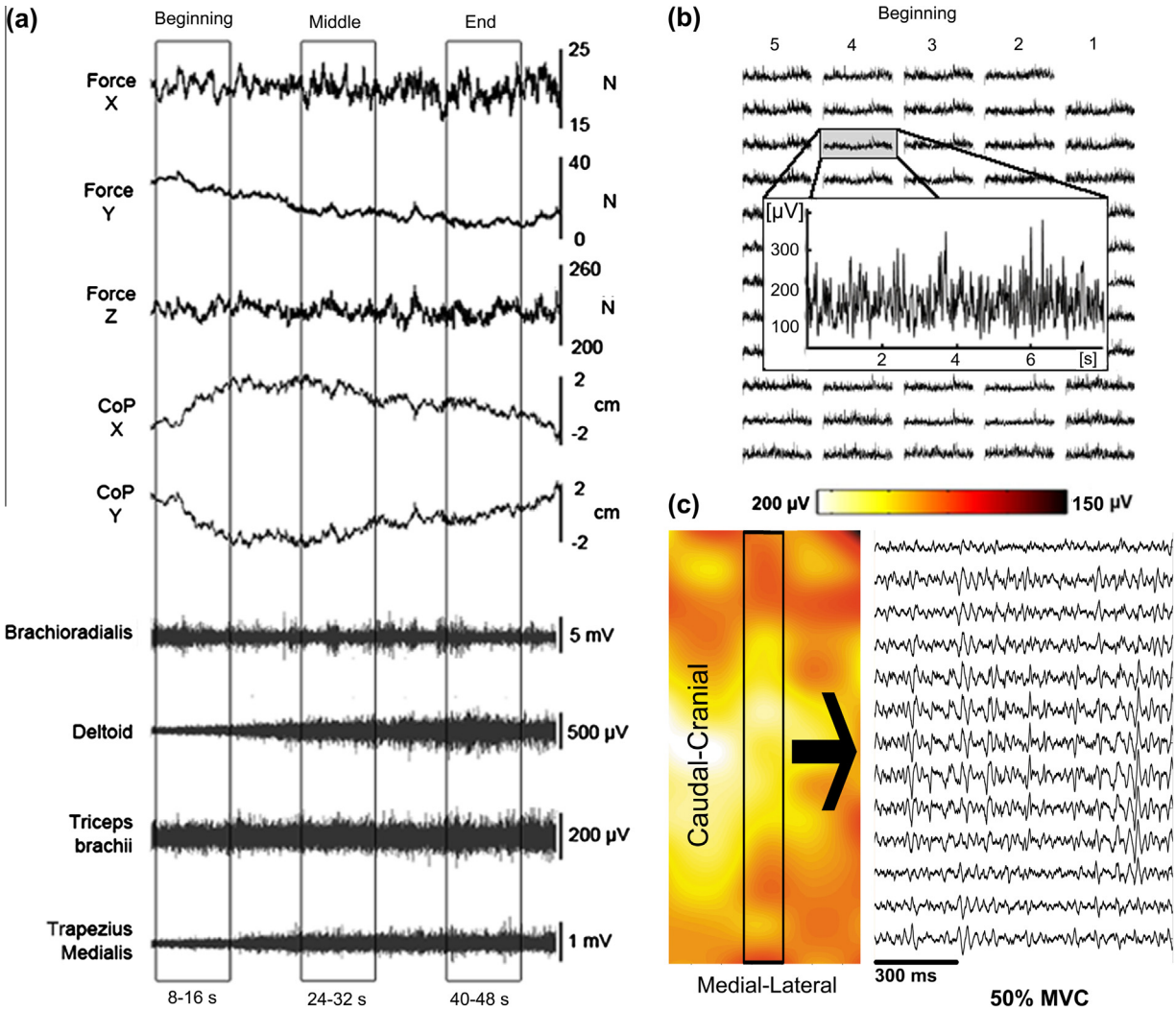

Fig. 2. Representative signals recorded during elbow flexion contractions at 50%MVC. (a) Force components, time series of centre of pressure (CoP), and bipolar EMG recordings from upper limb muscles. Analysis epochs (beginning, middle, and end) are highlighted together with the corresponding time period. (b) Full-wave rectified and low-pass filtered EMG activity recorded with the electrode grid from the biceps brachii muscle during the beginning epoch (8 s) at 50%MVC. (c) Topographical map of root mean square EMG in the beginning of the contraction for one subject. Next to the map, raw EMG signals from the middle column are displayed showing MUAP propagation. The topographical map was interpolated by a factor of 3 between each pair of electrodes for graphical representation.

3.1. Force variability

The CV of task-related (Fz) force showed a significant difference for the level of contraction (RM-ANOVA: $F = 7.09$, $P < 0.01$), indicating that the maximum CV was found at 50% MVC compared with 5%, 15%, and 30%MVC (NK: $P < 0.01$; Fig. 3a). The CoP of tangential forces revealed a significant interaction between level of contraction and time (RM-ANOVA: $F = 3.52$, $P < 0.05$; Fig. 3b). The CoP at 50%MVC was significantly increased compared with the CoP at 15% and 30%MVC for all epochs (NK: $P < 0.001$). Only the CoP at 30%MVC was progressively increasing during the contraction (NK: $P < 0.05$).

3.2. Bipolar EMG recordings and average activity of the EMG grid

The ANOVA of the RMS EMG from biceps brachii, trapezius, and triceps brachii revealed a significant interaction between contraction level and time (RM-ANOVA: $F > 2.32$, $P < 0.05$; Fig. 4a–c), showing that muscle activity was monotonically increased in the highest levels of contraction over time (50%MVC; NK: $P < 0.001$). Similarly, the RMS EMG from deltoid and brachioradialis muscles showed a monotonically increase over time during all contraction levels (RM-ANOVA: $F > 6.25$, $P < 0.05$, NK: $P < 0.001$; Fig. 4d and e).

The NMI between the task-related force component and the EMG recordings from biceps brachii, brachioradialis, deltoid, triceps brachii, and trapezius muscles decreased significantly over the epochs (RM-ANOVA: $F > 6.25$, $P < 0.05$; NK: $P < 0.001$; Fig. 4f).

3.3. RMS EMG and NMI maps distribution

An example of RMS EMG and NMI distributions on the grid during three epochs is illustrated in Fig. 5. Redistribution of muscle activity was observed over time, as shown by displacement of the RMS EMG map centroid. Similarly, changes in the NMI maps centroid position suggests that the information shared between the EMG and the task-related force variability are unevenly distributed in the muscle. Furthermore, significant differences were found between the RMS EMG and NMI centroid shifts in the medial–lateral direction (RM-ANOVA: $F = 8.51$, $P < 0.05$, NK: $P < 0.05$; Fig. 6), reflecting that more displacement of the centroid occurred in the RMS EMG distribution compared with the NMI centroid.

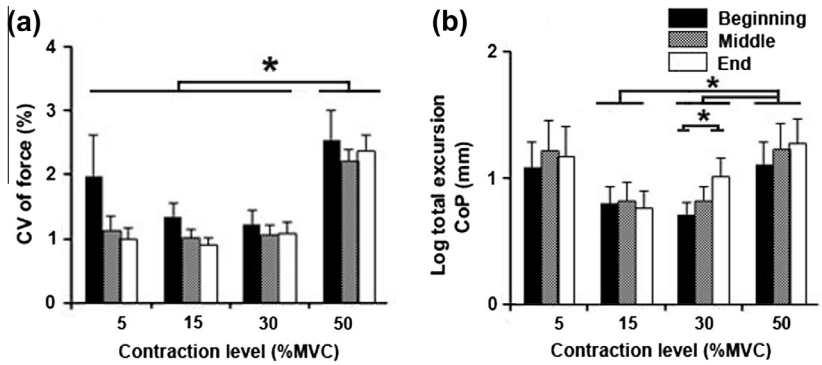

Fig. 3. Mean (+SEM, $N = 14$) coefficient of variation (CV) of force (a) and total excursion of centre of pressure (b) for contractions at 5%, 15%, 30%, and 50%MVC, during three different periods of the contractions: beginning (8–16 s), middle (23–31 s), and end (40–48s). $^{*}P < 0.05$ for multiple pair-wise comparisons using NK.

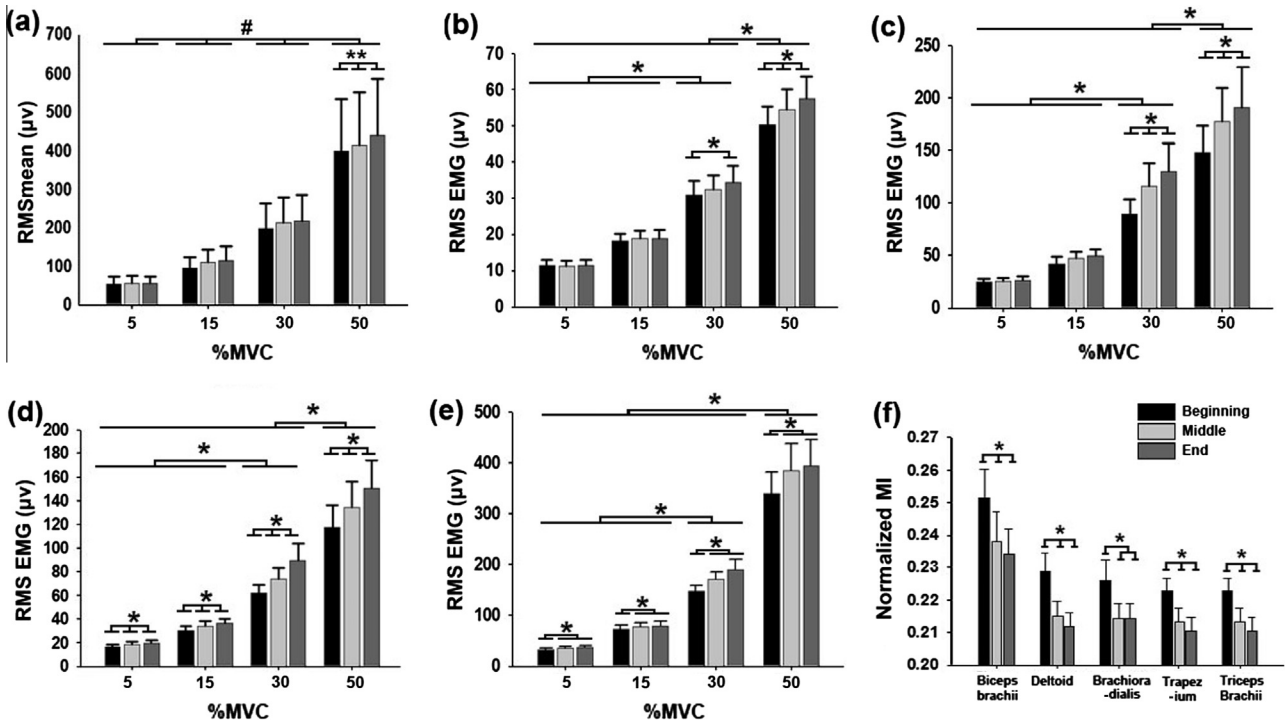

Fig. 4. Average (+SEM, $N = 14$) root mean square (RMS) EMG: (a) averaged over the EMG grid on the biceps brachii muscle, (b) triceps brachii, (c) trapezius medialis, (d) deltoid, (e) brachioradialis muscles at 5%, 15%, 30%, and 50%MVC, during three different periods of the contractions: beginning (8–16 s), middle (23–31 s), and end (40–48 s). (f) Normalised mutual information between task-related force and EMG signals showed a decrease over time for all signals. $^{*}P < 0.05$ for multiple pair-wise comparisons using NK.

3.4. Centroids excursion and its correlation with the tangential forces

In the medial–lateral direction, the difference between RMS EMG and NMI centroid shifts was positively correlated with the excursion of the CoP at 30% MVC at the end of the contraction ($R^2 = 0.35$, $P < 0.05$) and at 50% MVC during all the contraction periods ($R^2 = 0.39$, $R^2 = 0.33$, $R^2 = 0.30$, respectively; $P < 0.05$).

4. Discussion

This study assessed the spatial distribution of muscle activity and its contribution to the force variability across time using linear and nonlinear techniques. The results showed a difference in the evolution of the RMS EMG and NMI spatial organisation during the sustained contractions, calculated by changes of position of the centroids in two time periods. The difference between centroids position was correlated with variability in tangential forces at 30% and 50% MVC, reflecting an interaction between muscle adaptations to tangential force variability.

4.1. Spatial redistribution of muscle activity

Central and peripheral mechanisms are involved in the spatial adaptation of muscle activity during sustained contractions (Farina et al., 2008). Central mechanisms modulate the motor unit activity by an increase in the central drive and by recruiting/derecruiting motor units (Farina et al., 2008; Gandevia, 2001). The peripheral mechanisms involve changes in the fibre membrane properties such as non-uniform decrease of the muscle fibre conduction velocity (Merletti et al., 1990) and modulations of the intracellular action potential shapes (Dimitrova and Dimitrov, 2003). In the present study, although the average RMS EMG value increased,

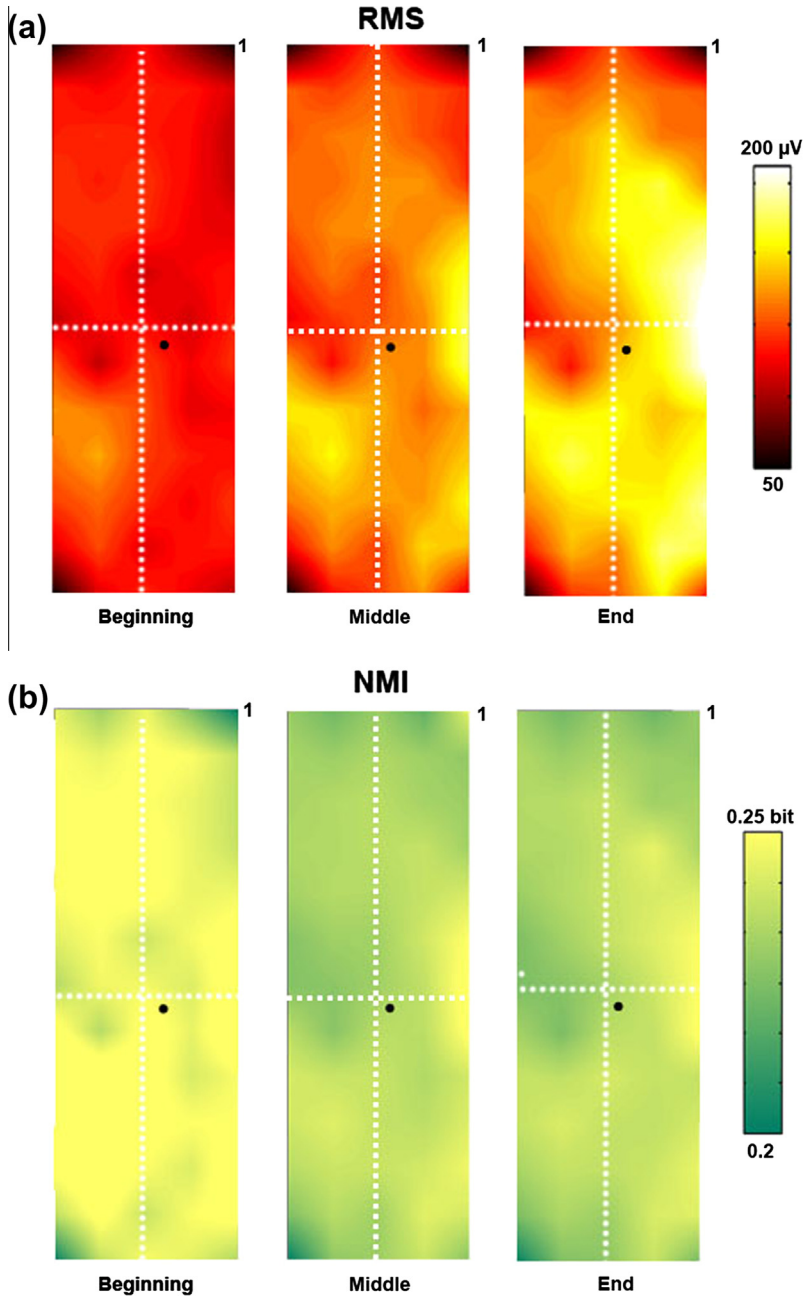

Fig. 5. Representative example of the NMI between task-related force and RMS EMG over the grid from 8 s-long signal epoch. The RMS EMG and NMI are represented during the beginning of the contraction (8–16 s), middle (24–32 s), and to the end of the contraction (40–48 s) at the 30% MVC. The position of the centroid is illustrated by a black dot. (a) Root mean square EMG distribution within the grid. (b) Mutual information between EMG in the grid and task-related force. White dashed lines represent the centre of the grid, and the grid's corner in the lateral-cranial position is showed by a 1.

the increase in muscle activity was unevenly distributed during the sustained contractions, as reflected by changes in the RMS EMG centroid position.

Heterogeneous spatial muscle adaptations were observed previously, and two hypotheses has been proposed to explained this phenomenon: confined location of most fatigable motor units in different regions, or dissimilar increment of the activity of subvolumes within a muscle (Gallina et al., 2011). Nevertheless, literature have shown that fibres type II are the muscle fibres most susceptible to fatigue (Fitts, 1994), and histologic results have revealed that fibres type II are homogeneous distributed in the cross-section of the muscle (Dahmane et al., 2005). Therefore, in agreement with the previous study, significant difference over time would most likely be found in the direction of the muscle fibre (cranial-caudal direction) rather than in the medial–lateral direction, if the muscle adaptations demonstrated in the present study were cause by change in the membrane properties. Regarding activity of muscle subvolumes, dynamic reorganisation of the muscle activity within the muscle during sustained contractions has been described (Holtermann et al., 2010, 2008). Moreover, it has been reported that an EMG increase is related with the number of newly recruited motor units during sustained low-level contractions (Fallentin et al., 1993), supporting the stronger impact on force variability by additional motor unit recruitment (Contessa et al., 2009).

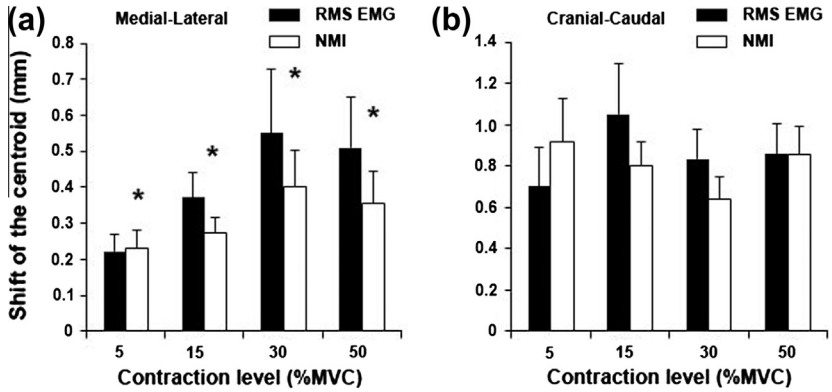

Fig. 6. (a) Shift of the RMS EMG and NMI centroids of the grid in the medial–lateral direction, representing absolute displacement of the centroid in maps between the beginning and the end of contraction periods for contractions at 5%, 15%, 30%, and 50%MVC. Shifts in RMS EMG centroid position were significantly higher compared with NMI centroid position (*$P < 0.05$). (b) Shift of the RMS EMG and NMI centroids of the grid in the cranial–caudal direction.

4.2. Relationship between force variability and EMG activity

To date, linear correlation has been the most common methodology applied to quantify the relationship between superficial EMG activity and force output (Maier and Hepp-Reymond, 1995; Metral and Cassar, 1981). The method assumes a linear relationship between muscle activity and force production. However, several factors can modulate the superficial EMG morphology without a real variation of the muscle activity (Farina et al., 2004). The modulation of the EMG yields to a lower correlation between muscle activity and force output during sustained contractions. Thus, previous studies have tried to improve the linear correlation by removing low frequency components of EMG signals (Potvin and Brown, 2004; Staudenmann et al., 2006), reducing common components in the EMG (Farina et al., 2004; Potvin and Brown, 2004; Staudenmann et al., 2006).

An alternative to the linear correlation is the MI. The method can identify both linear and nonlinear dependencies between signals (Madeleine et al., 2011), regardless of the relationship between the EMG and the force variability. The literature describes the MI as a suitable approach when the relationship between two signals is complex, and it is generally defined as a quantity to describe the correlation or similarity between signals (Li, 1990). Previous studies have used MI on EMG signals to quantify functional connectivity between muscle-subdivision (Madeleine et al., 2011), between different muscles (Svendsen et al., 2011), or to describe dependency between deltoid muscle activity and arm movements (Farfán et al., 2010). In the present study, NMI was used for the first time to quantify the spatial contribution within the biceps brachii muscle activity on the force variability during elbow flexion contractions. The results showed an uneven distribution of the information shared between EMG and the force variability across the grid, suggesting that some muscle regions may contribute differently to force variability during sustained contractions. However, most likely this heterogeneity in the NMI maps is cause by anatomical or volume-conductor-related differences affecting the EMG recordings.

Interestingly, the shifts of the EMG RMS and the NMI centroids were slightly different, implying that some muscle regions can increase or decrease their EMG activity without necessarily increasing or decreasing the information shared with the task-related force variability over time. Previously studies observed that different spatial EMG patterns occur during different motor tasks, reflecting a regional organisation of the muscle (Staudenmann et al., 2009). Depending on the fatigue resistance of the muscle regions, the EMG patterns could change over time, and consequently affecting the muscle region contributions to the output force. This is in accordance with previous findings that described a dynamic load sharing within a muscle during long contractions (Holtermann et al., 2008) and spatially localised effects of fatigue on muscle activity (Gallina et al., 2011). According to recent studies, the ability to modulate the heterogeneity of the muscle activity is associated with a higher resistance to muscle fatigue (Farina et al., 2008), and could be part of a fatigue adaptation strategy. This is also consistent with other studies that showed no significant association between force variability and motor unit discharge rates variability (Macefield et al., 2000) or motor unit synchronisation (Farina et al., 2004; Semmler et al., 2000). In fact, it has been established that no single muscle adaptation mechanism can explained all changes on force variability (Taylor et al., 2003).

4.3. Modulation of the force variability

The global decrease of information between muscle activity recordings and task-related force over time could indicate that more muscles contribute to force variability during sustained contractions. Previous studies have highlighted the role of muscle coordination on motor control accuracy during fatigue (Gribble et al., 2003; Missenard et al., 2008). Moreover, it has been suggested that less fatigued accessory muscles could compensate for motor control deficiency (Poortvliet et al., 2013; Rudroff et al., 2007). In this regard, increased muscle cooperation could enlarge the force variability, since a cooperative muscle would contribute to force on its direction of mechanical action (Kutch et al., 2008). The resultant summation of all the individual muscle contractions would reflect a portion not only of the task-related force variability (Svendsen et al., 2011), but also of the tangential force variability (Salomoni and Graven-Nielsen, 2012a; Svendsen and Madeleine, 2010).

During the sustained contractions in the present study, not only the primary muscle involved in the elbow flexion contractions increase their activity, but also antagonist and stabiliser muscles monotonically raised their activity supporting the primary mover. The change in the distribution of the primary mover activity has been related to muscle fatigue resistance (Farina et al., 2008). However, co-contractions of antagonistic muscles in the human arm can be activated or inhibited in parts of the motoneurons pools of antagonistic (Jongen et al., 1989). This phenomenon could play a role in the inhomogeneous activity of the primary muscle. Therefore, changes in the activity of the supporting muscles during the contraction can impact on the motoneuron pool on the primary muscle, altering the motor unit activity and consequently, the force output direction (Tucker and Hodges, 2010). However, it is not

possible to quantify motor units with the current assessment methods (Farina et al., 2004).

Previous studies have reported modulation of force variability during muscle fatigue (Contessa et al., 2009; Salomoni and Graven-Nielsen, 2012a). In the present results, the motor system was challenged to sustain long contractions, but it is unlikely that higher levels of muscle fatigue were developed. Interestingly, tangential force variability was increased over time at 30% of maximal contraction, and this could be associated with increased auxiliary muscles activity without increasing the biceps brachii activity. However, the remaining levels of contractions did not reveal any change over time. The discrepancy could be related to the insufficient contractions time required, inducing low muscle adaptations. Another possible explanation is that the motor system was able to compensate muscle fatigue sustaining the precision using alternative strategies (Gates and Dingwell, 2010).

Even though the force variability did not change significantly over time, a correlation between the tangential force variability and the difference between the centroids (EMG and NMI) were found. This reflects that the difference between the muscle activation and its contribution on the task-related force variability also play a role in the modulation of the tangential forces variability. It is suggested that the motor system compensate the motor unit inhomogeneous activity of primary movers increasing the activity of additional muscles, affecting the force variability in different directions. Moreover, previous studies have highlighted the increased activity in muscles not contributing on the external torques in complex motor tasks (Flanders and Soechting, 1990; Zuylen et al., 1988). Therefore, most likely the centroid difference is compensated by the increased activity of the auxiliary muscle, and it is mainly the auxiliary muscles which account for most of the tangential force variability (Poortvliet et al., 2013; Rudroff et al., 2007).

Two reasons have been suggested to explain the purpose of compensation by auxiliary muscle. First, increased auxiliary muscle activity aimed to balance the relative contribution of muscles acting on the task-related force direction as well as on unwanted directions (Zuylen et al., 1988). Second, coactivation of auxiliary muscles might be a mechanism to counteract internal forces and protect the joint (Flanders and Soechting, 1990). In the present study, dissimilarity between muscle activity and its contribution to the force output cause forces in unwanted directions. Thus, it is necessary to compensate deviation from the target force in order to have more precision in the contractions and to avoid joint overload during sustained contractions.

5. Conclusion

The muscle activity contribution to force variability was heterogeneously observed over different parts of the muscle during the contractions, suggesting that muscle adaptations do not necessarily affect force variability. The contribution of muscle reorganisation to force variability was quantified by assessing shifts of the RMS EMG and the NMI centroids. Interestingly, the centroids' difference was correlated with the CoP of the tangential force, implying that muscle adaptations affect tangential force variability.

Conflict of interest

The authors have no conflict of interest for this study.

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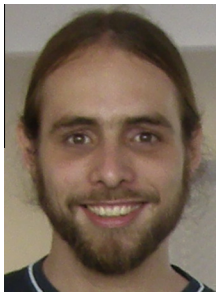
Christian Ariel Mista graduated as bioengineer in 2010 at the Faculty of Engineering of the National University of Entre Ríos, Argentina. He is currently employed as Ph.D. student at the Center for Sensory–Motor Interaction (SMI), Department of Health Science and Technology at Aalborg University, Denmark. His main areas of research are biomedical signal processing with focus on multichannel surface electromyography and force variability.



Sauro Emerick Salomoni received his M.Sc. in Electric Engineering (University of Brasília, Brazil) in 2008 and his Ph.D. within Biomedical Science and Engineering in 2012 (Aalborg University, Denmark). Currently he is postdoctoral fellow at NHMRC Centre of Clinical Research Excellence in Spinal Pain, Injury and Health, The University of Queensland, Brisbane, Australia. His research interests focus on motor adaptations during experimental and chronic pain conditions.

Thomas Graven-Nielsen received his Ph.D. within Biomedical Science and Engineering in 1997 (Aalborg University, Denmark). In 2006 he obtained a Doctoral degree in Medical Science (DMSc, Copenhagen University, Denmark). He is Full Professor in Pain Neuroscience at Center for Sensory-Motor Interaction, Department of Health Science and Technology, Aalborg University. The core research area is motor control and pain, muscle pain, referred pain, deep-tissue hyperalgesia, and electrophysiological techniques to assess muscle pain physiology.